Number 1 — Fractions · Order of operations · Rounding

No calculator. Give answers as fractions or mixed numbers in their lowest terms unless the question says otherwise. Marks are shown in brackets. ★ = harder question (grades 7–9).

A · FRACTIONS

1 Simplify [3]

a $\frac{18}{48}$ **b** $\frac{150}{48}$ **c** $\frac{2.4}{48}$

2 Work out [4]

a $\frac{4}{9} \times 1\frac{1}{8}$ **b** $3\frac{1}{3} \div \frac{5}{6}$

3 Work out $\frac{5}{6} + 2\frac{3}{4} - 1\frac{2}{3}$ [3]

4 In a school, $\frac{2}{5}$ of the students study Spanish, $\frac{1}{4}$ study French, $\frac{1}{10}$ study German and the rest study no language. [5]

a What fraction of the students study no language?

b 90 students study no language. How many students are in the school?

5 A jug holds $3\frac{3}{4}$ litres of juice. A cup holds $\frac{3}{8}$ litre. How many full cups can be poured from the jug? [3]

B · ORDER OF OPERATIONS

6 Work out [4]

a $5 + 3 \times (2 + 3)^2$ **b** $\frac{2}{3} + \frac{1}{3} \div \frac{4}{9}$

7 Insert one pair of brackets to make this calculation correct: $2 + 3 \times 4 - 1 = 19$ [2]

8 Evaluate $\left(1\frac{1}{2} - \frac{2}{3}\right)^2 \div \left(\frac{5}{6} + \frac{1}{4}\right)$ [3]

C · SIGNIFICANT FIGURES AND DECIMAL PLACES

9 The height of Mount Everest is $8\,848.86$ m. Write this height correct to [2]

a 1 decimal place **b** 2 significant figures

10 $\pi = 3.14159265\ldots$ Write π correct to [2]

a 4 significant figures **b** 4 decimal places

11 The mass of a grain of pollen is $0.000\,004\,09$ g. [3]

a Write this mass correct to 1 significant figure.

b Write it correct to 6 decimal places.

c Explain why your answer to **b** is less useful than your answer to **a**.

D · MIXED PROBLEMS

12 A bag of rice weighs $2\frac{2}{5}$ kg. [5]

a A café uses $\frac{3}{10}$ kg of rice each day. For how many full days does the bag last?

b A different café uses $\frac{1}{3}$ of a new bag in the first week and then $\frac{3}{4}$ of what is left in the second week. What mass of rice is left after two weeks?

13 Kai says: "-3^2 and $(-3)^2$ are the same because both contain -3 and a square." Is Kai right? Work out both values and explain. [3]

Total: 42 marks

Check: fractions in lowest terms · mixed numbers where the answer is bigger than 1 · units in word problems.